UNIVERSITY of MARYLAND
MARLENE AND STEWART GREENEBAUM
COMPREHENSIVE CANCER CENTER

ISAC RECOGNIZED
Shared Resource Laboratory

“Deciphering the black box”: leveraging shared-resource laboratory resources to tackle spectral flow cytometry’s quirks

David Rach, Ph.D.
UMGCCC Flow Cytometry Shared Resource
Baltimore, MD

November 05, 2026

Slides

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Spectral Flow Cytometry is amazing!!!

- Capable of resolving remarkably similar fluorophores → **increased number of markers per panel** compared to conventional flow cytometry (CFC)
- Allows for rapid acquisition of millions of events in a few minutes → typically **greater number of acquired cells per sample** compared to mass cytometry (MC)
- Spectral Flow Cytometry (SFC) datasets have an unique combination of **width and depth** compared to preceding technologies



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Spectral Flow Cytometry is amazing!!!



- Able to efficiently acquire greater number of cells → carry out a more thorough profiling of **rare cell subsets**
- Build one panel, evaluate multiple cell types → invaluable for clinical studies with **limited biospecimen**
- Expand our knowledge about immune system heterogeneity

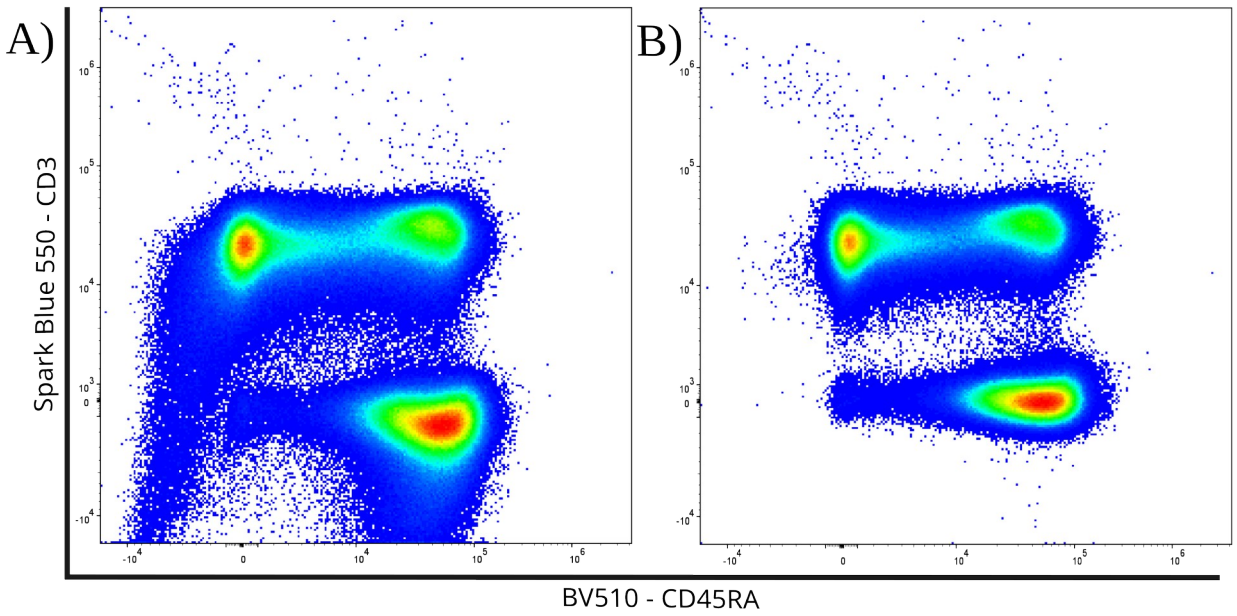
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Adapted algorithmic tools struggle with unmixing errors

- Adapted software packages heavily rely on having a consistent **median fluorescent intensity (MFI)** to cluster cells based on positive or negative marker expression

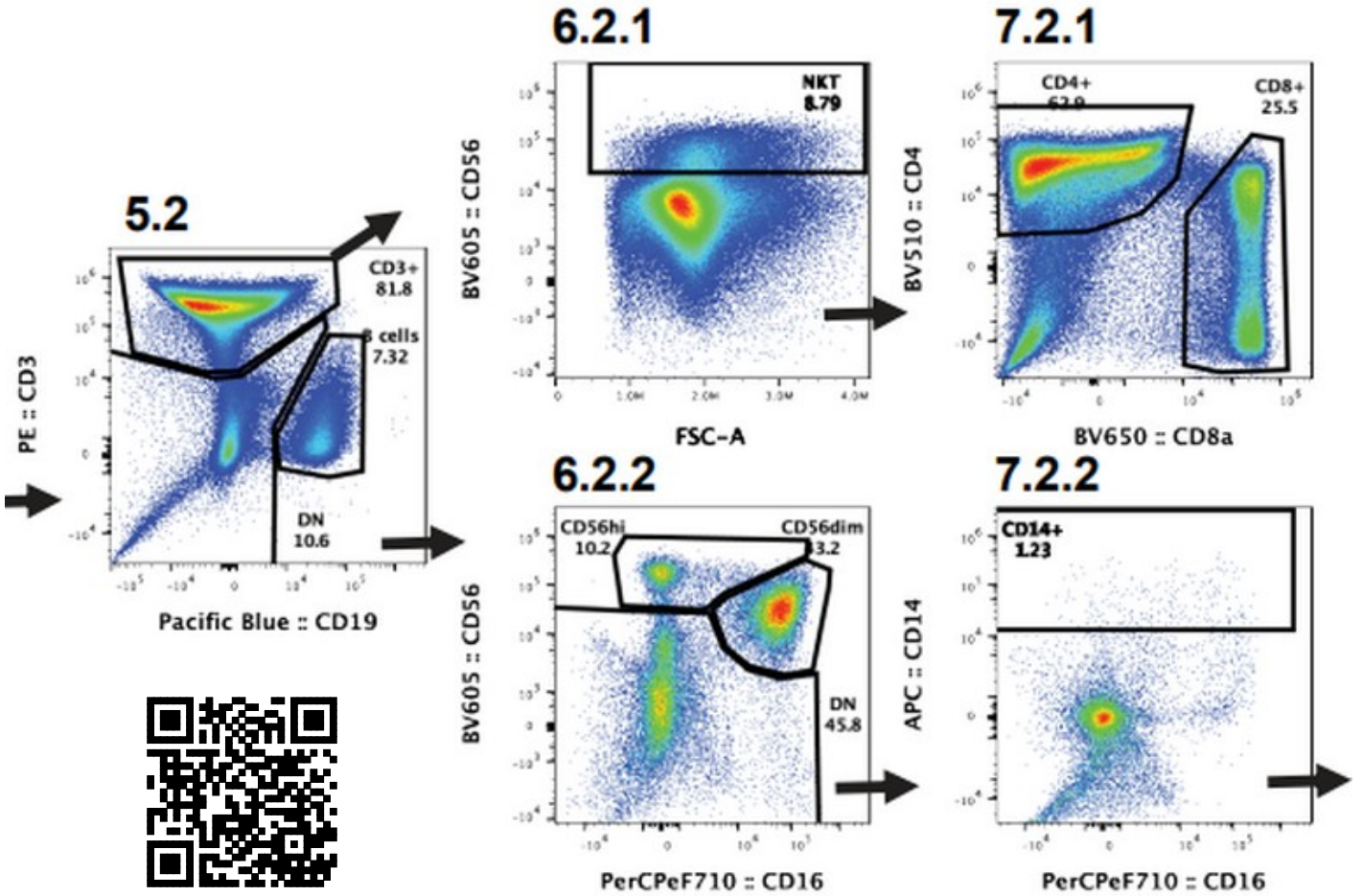
- Existing **batch correction** / normalization / scaling tools target instrumental or **experimental** level variation for correction

- **Unmixing errors** can seriously shift the MFI ranges for “negative” and “positive” cells for **individual** acquired specimens



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“We discovered a novel dendritic cell cluster”



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Building open-source tool-sets to address SFC's quirks



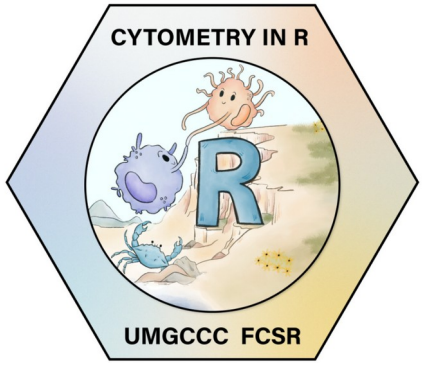
Spectral Signature Characterization and Isolation

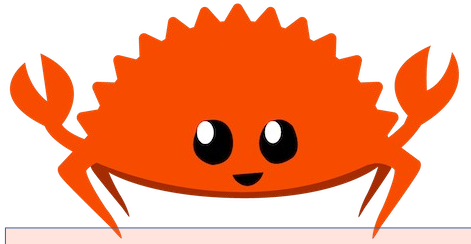


Incorporation of manually-annotated gating information



Automated Instrument Quality-Control Dashboards

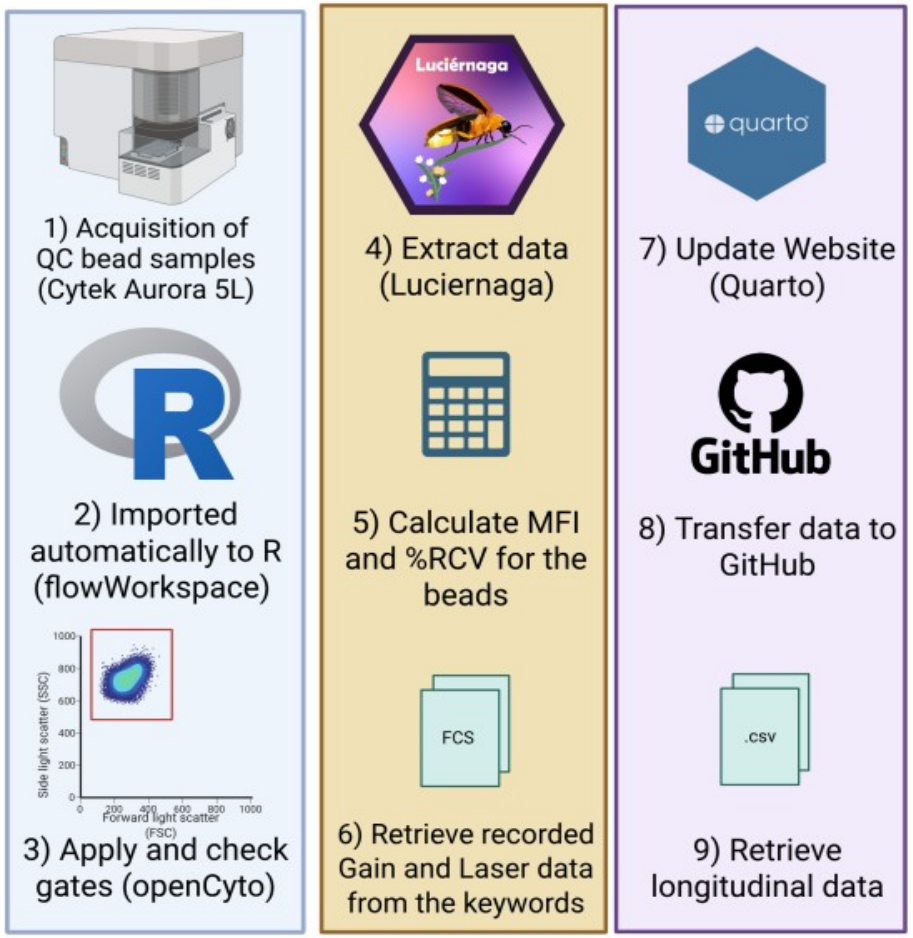




Rust

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Automated Daily Processing





Behind-the-scenes:

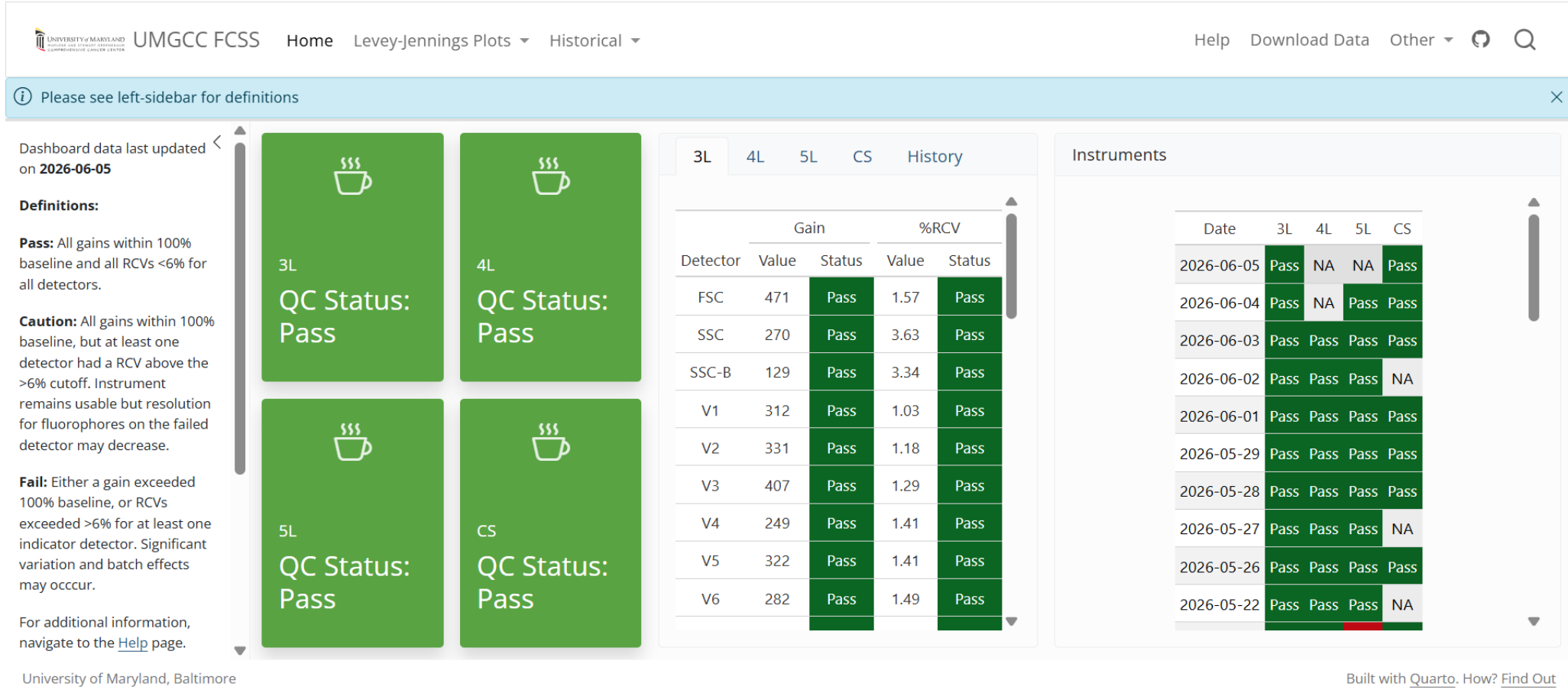
- Using R (and increasingly Rust) to process the .fcs files
- Automated transfer to GitHub via Windows Task Scheduler
- Our Quarto Website gets updated twice daily via a GitHub action

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Instrument Quality Control Dashboard

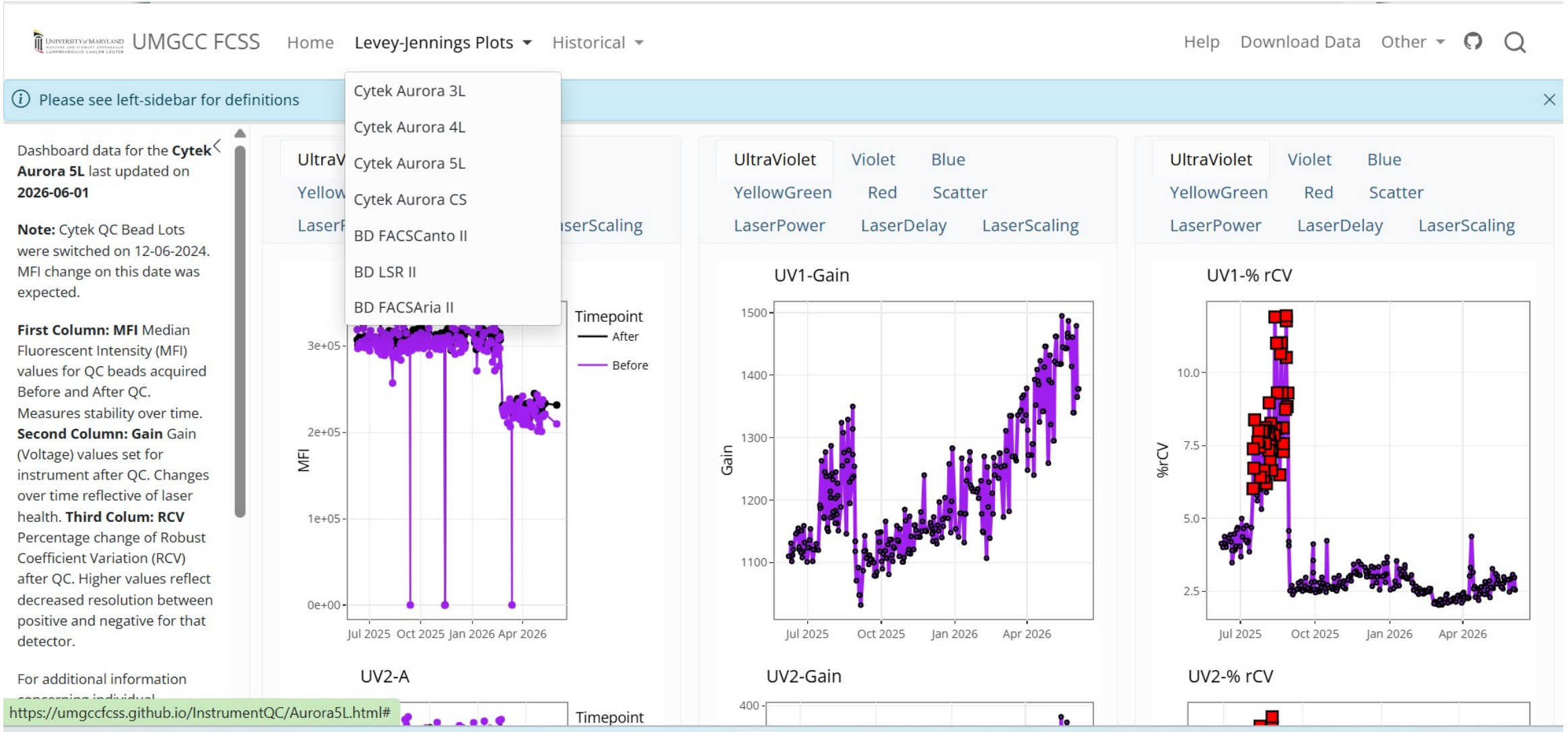
Dashboard

How-To



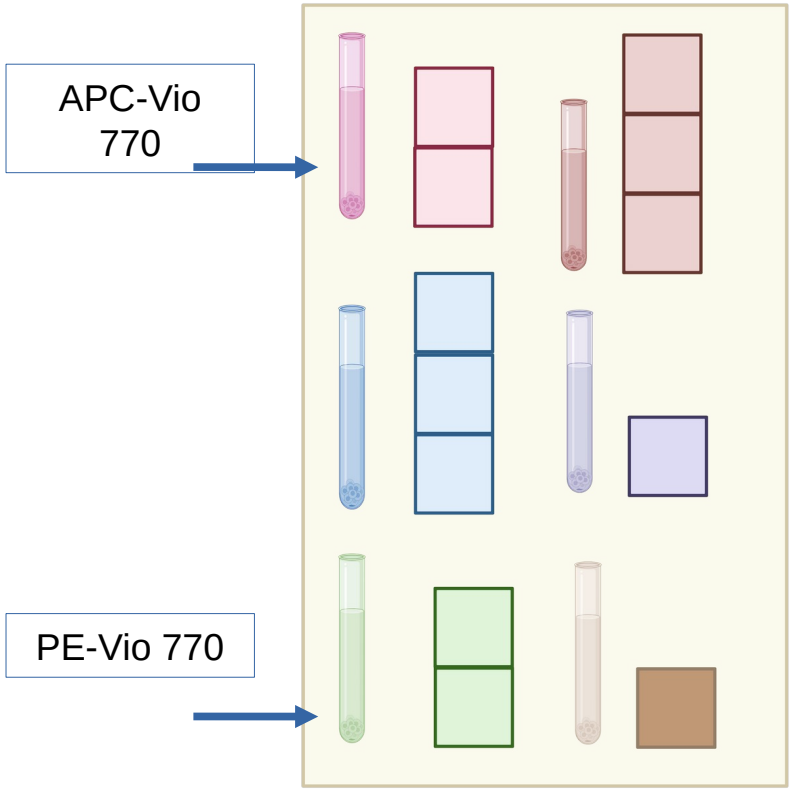
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Instrument Quality Control Dashboard

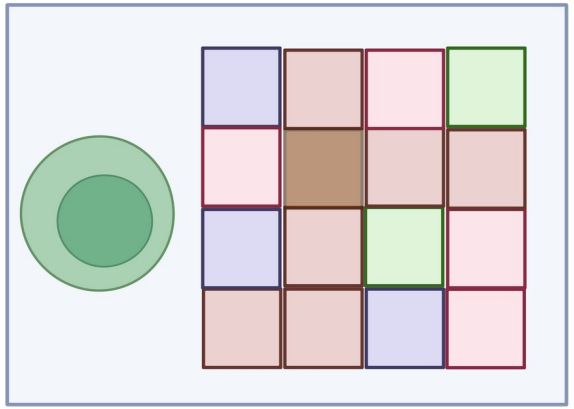


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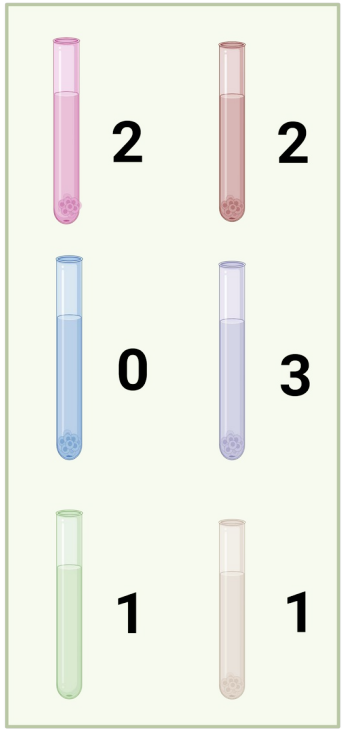
Spectral Unmixing



Unmixing Controls (Single Color and Unstained)



Single cell from full-stained sample



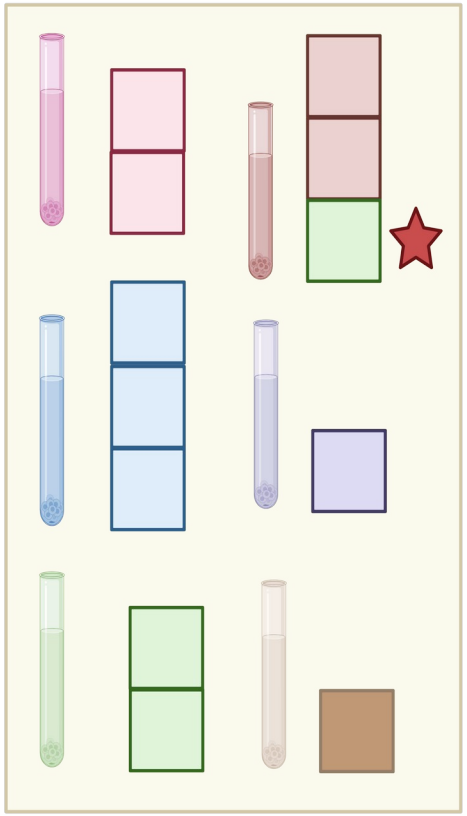
What fluorophores are present, and roughly how much

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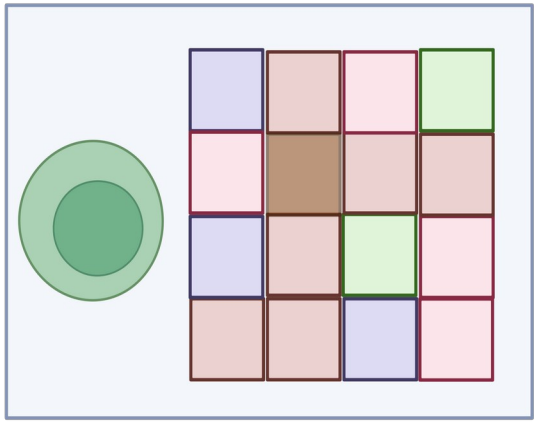
Mismatched references lead to errors in attribution

APC-Vio 770

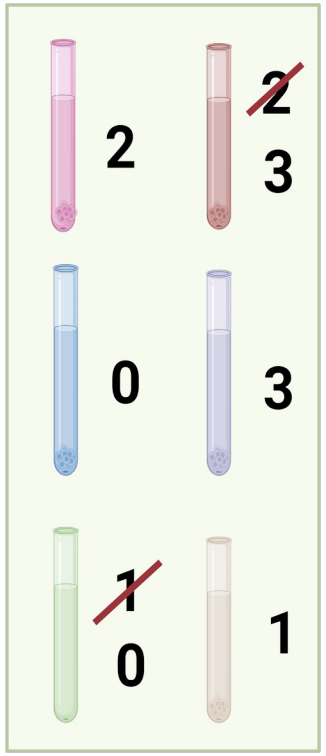
PE-Vio 770



Unmixing Controls (Single Color and Unstained)



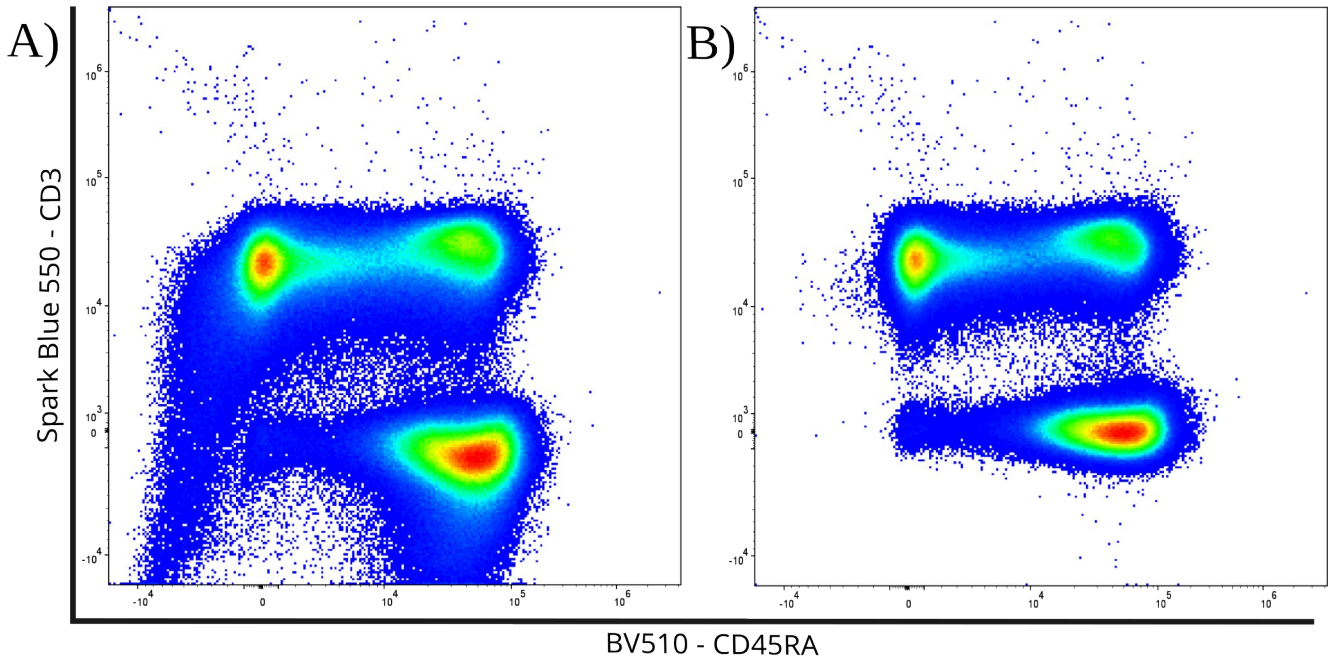
Single cell from full-stained sample



What fluorophores are present, and roughly how much

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How to identify problematic unmixing controls



- Despite the critical importance unmixing controls play, few tools permit us to determine quality of the individual control before unmixing

- Existing quality control algorithms (PeacoQC, FlowAI, etc.) primarily focus on detecting laser and fluidic issues in already acquired .fcs files.

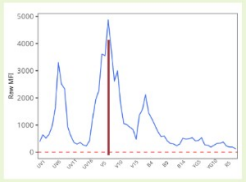
- Unmixing errors can be introduced by tandem degradational, additional autofluorescences, undiagnosed instrumental issues, etc.

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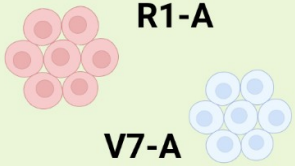
Luciernaga



Code



Identify detector with brightest MFI for each cell



Group cells with shared peak detectors



Subtract Out Autofluorescence

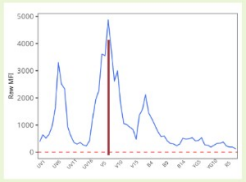
University of Maryland School of Medicine

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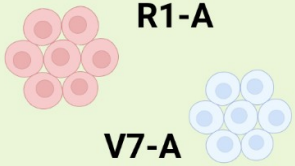
Luciernaga



Code



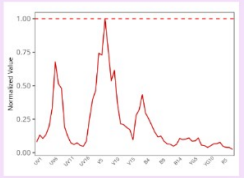
Identify detector with brightest MFI for each cell



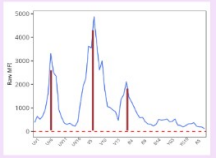
Group cells with shared peak detectors



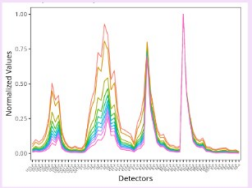
Subtract Out Autofluorescence



Normalize individual cell fluorescent signatures



Identify peaks using local maxima



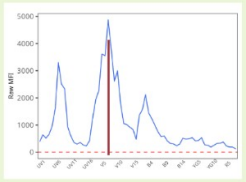
Group cells with shared peaks & heights

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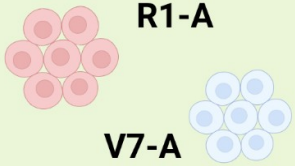
Luciernaga



Code



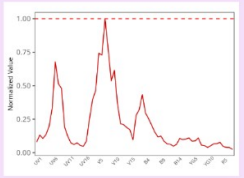
Identify detector with brightest MFI for each cell



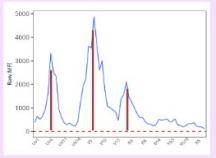
Group cells with shared peak detectors



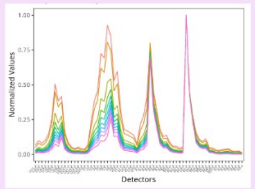
Subtract Out Autofluorescence



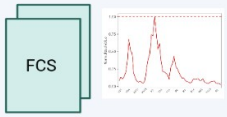
Normalize individual cell fluorescent signatures



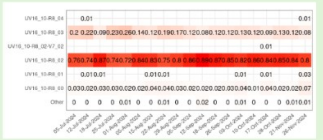
Identify peaks using local maxima



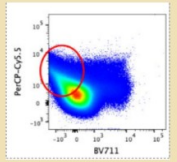
Group cells with shared peaks & heights



Enumerate and export out cells that share signature as .fcs files



Checks for tandem degradation

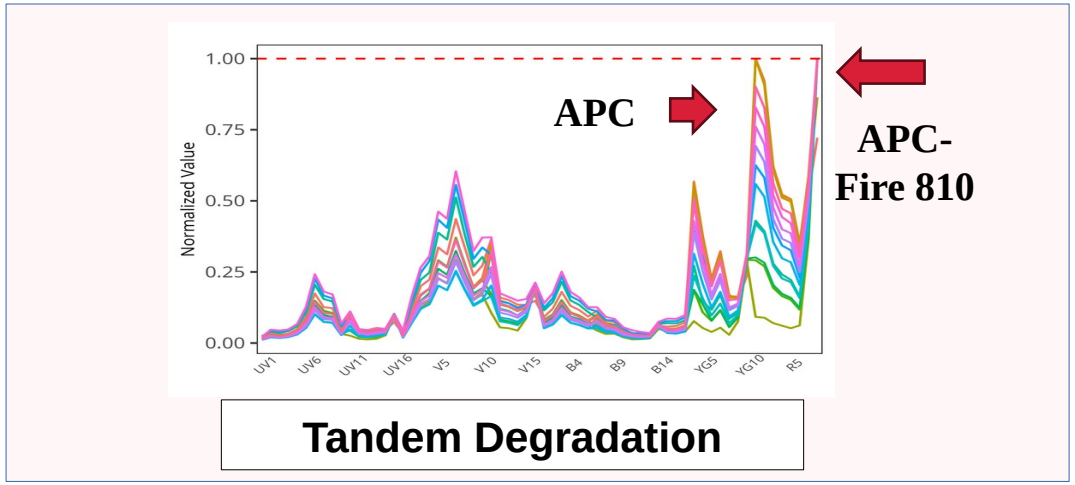


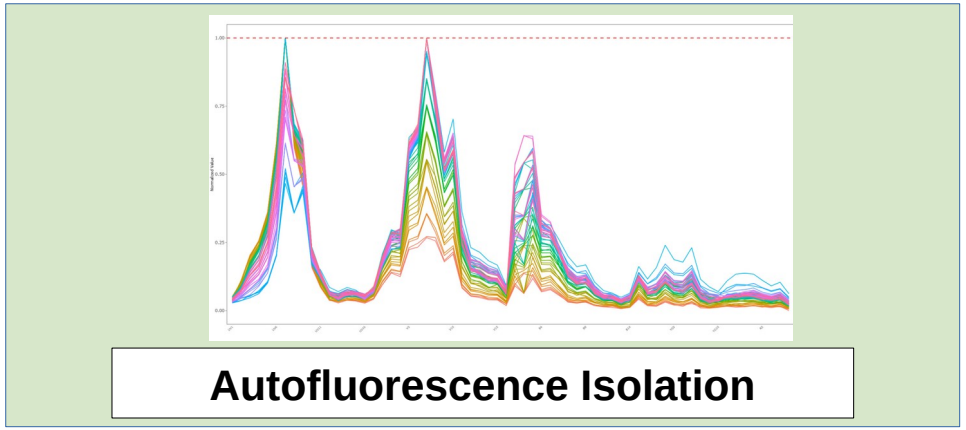
Autofluorescence Isolation

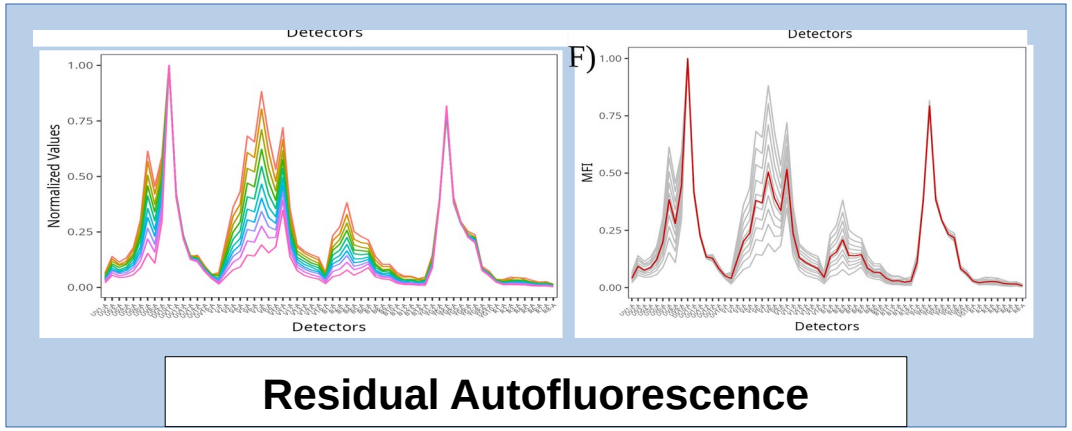
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Automated tools for screening unmixing controls







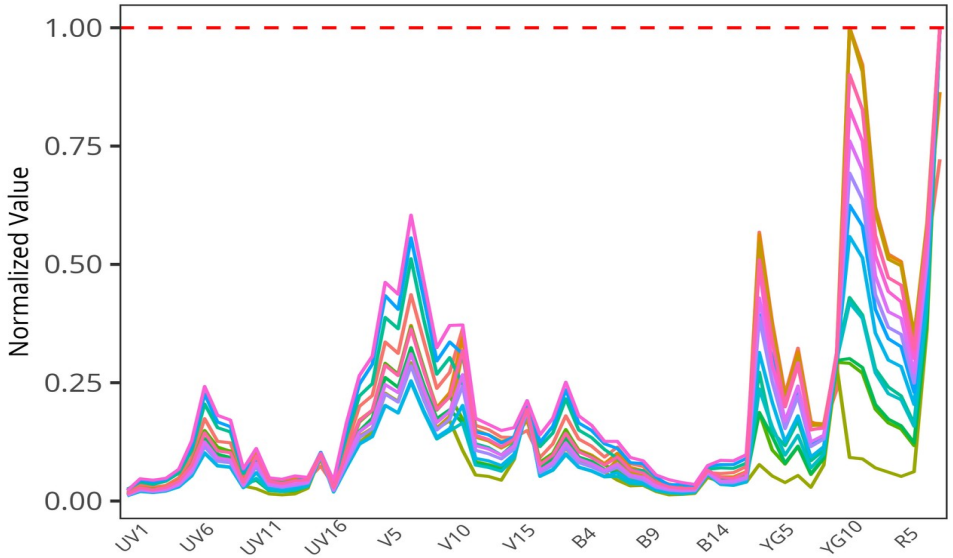
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Identifying degrading fluorophores

Fluorophores in good shape → **one predominant normalized signature** within the unmixing control raw .fcs file

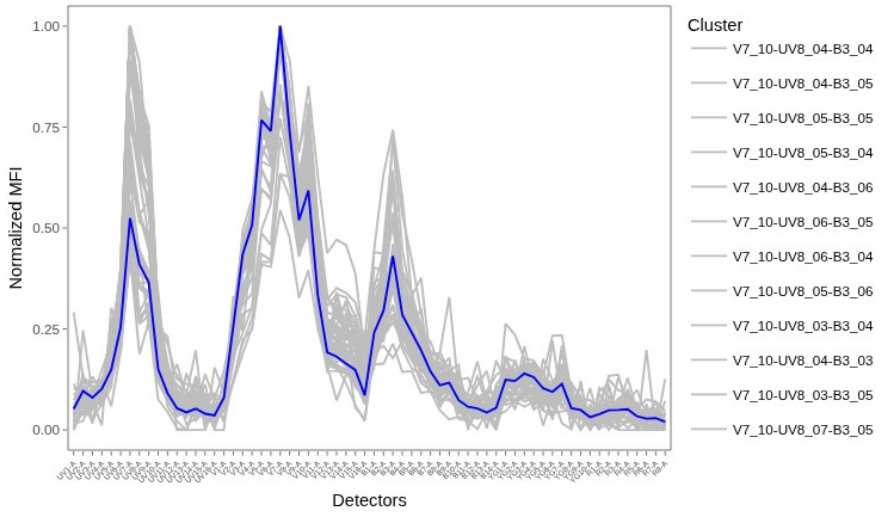
- When the fluorophore degrades, noticeable **increase in variant signatures**, making up substantial proportion of all cells in the unmixing control

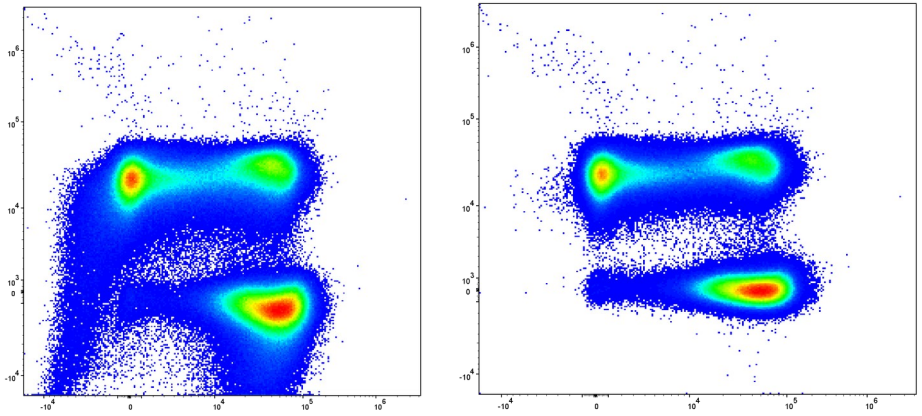
- Automated screening all acquired unmixing controls from campus users during off-hours



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Isolating additional autofluorescence signatures





- This approach also works to isolate variant autofluorescence signatures from unstained controls

- Unlike other approaches, the isolated cells share normalized signatures, only differing in terms of brightness

- Return directly as an .fcs file containing just cells with same signature for use as an autofluorescent tag in SpectroFlo

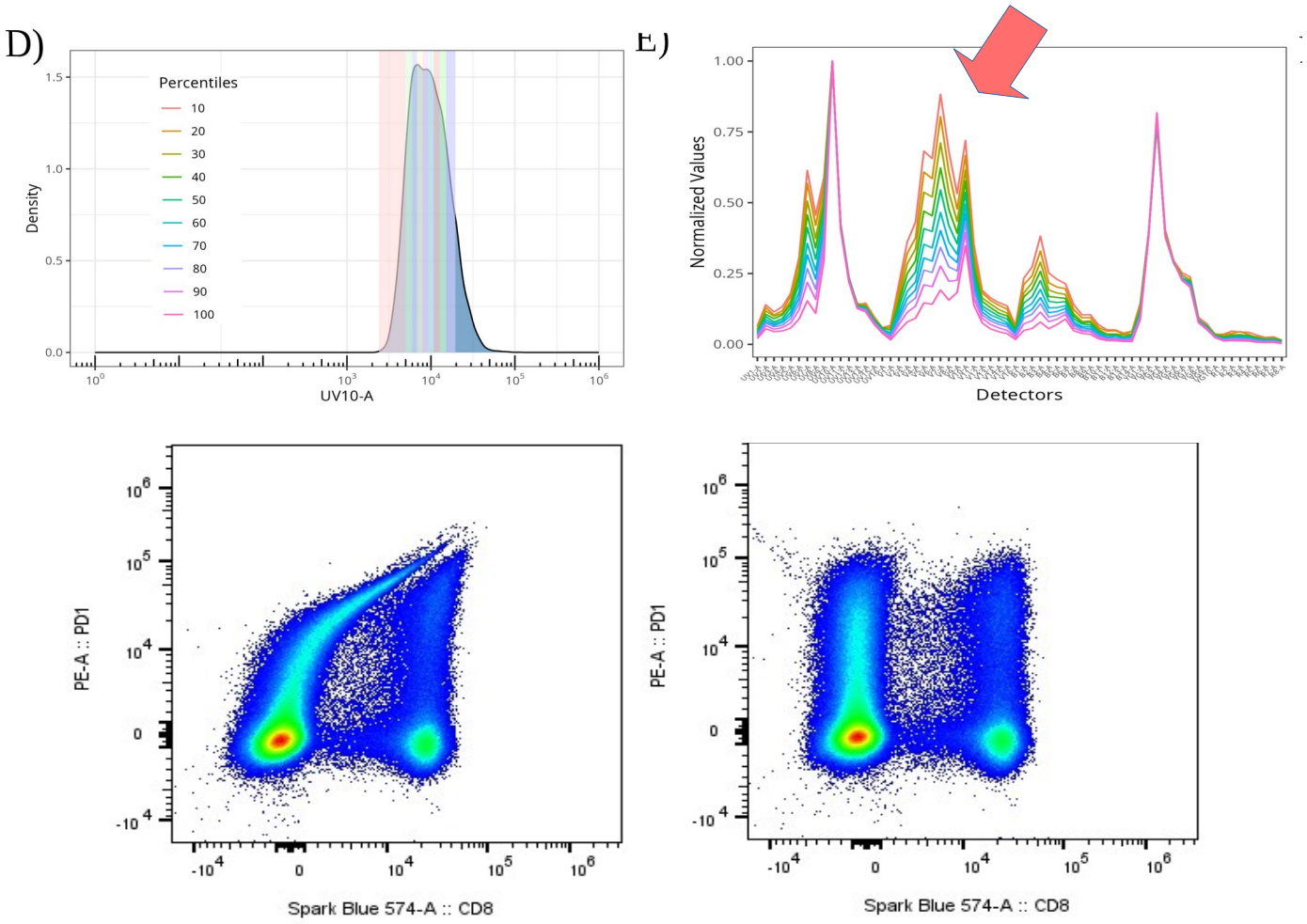
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Residual autofluorescence in dim cell single-color controls

- **“As bright or brighter”**

- Dim antigen-fluorophore combinations acquired as **cell single-color controls** often have residual autofluorescence contribution to their signature

- Identifying and **switching these** dim cell single-color controls to **bead single-color controls** significantly improved unmixing for our panels



Signature with residual AF

Signature without residual AF

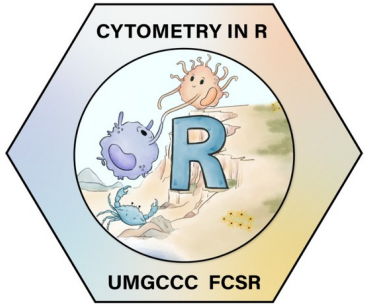
What good is any tool if the community is not able to use it?

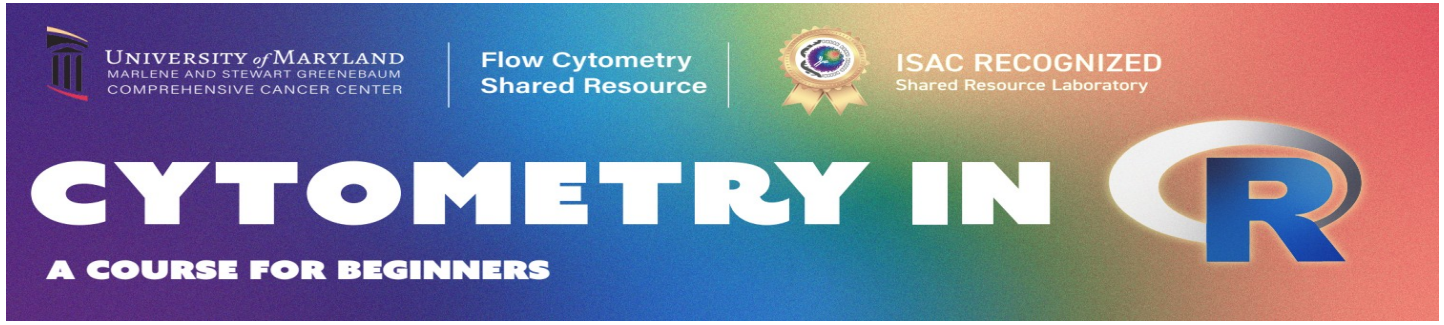
- Few resources to learn Cytometry in R to begin with
- Those that exist aimed at those with intermediate-advanced R skills, not beginners
- Know-how concentrated at a few institutions
- Most self-learning attempts end in frustration



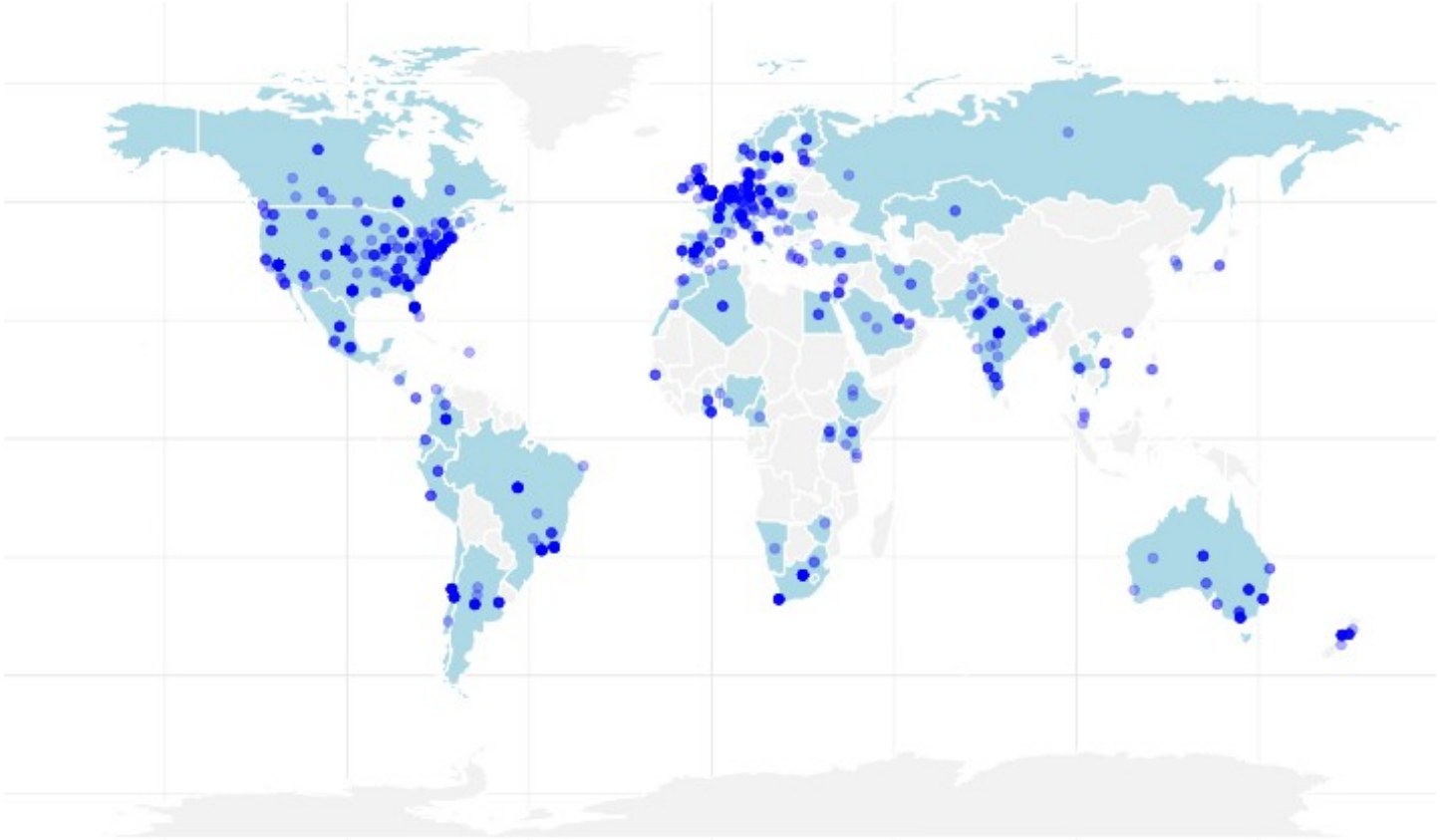
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Cytometry in R: A free weekly course for flow cytometrist with no-to-little coding experience





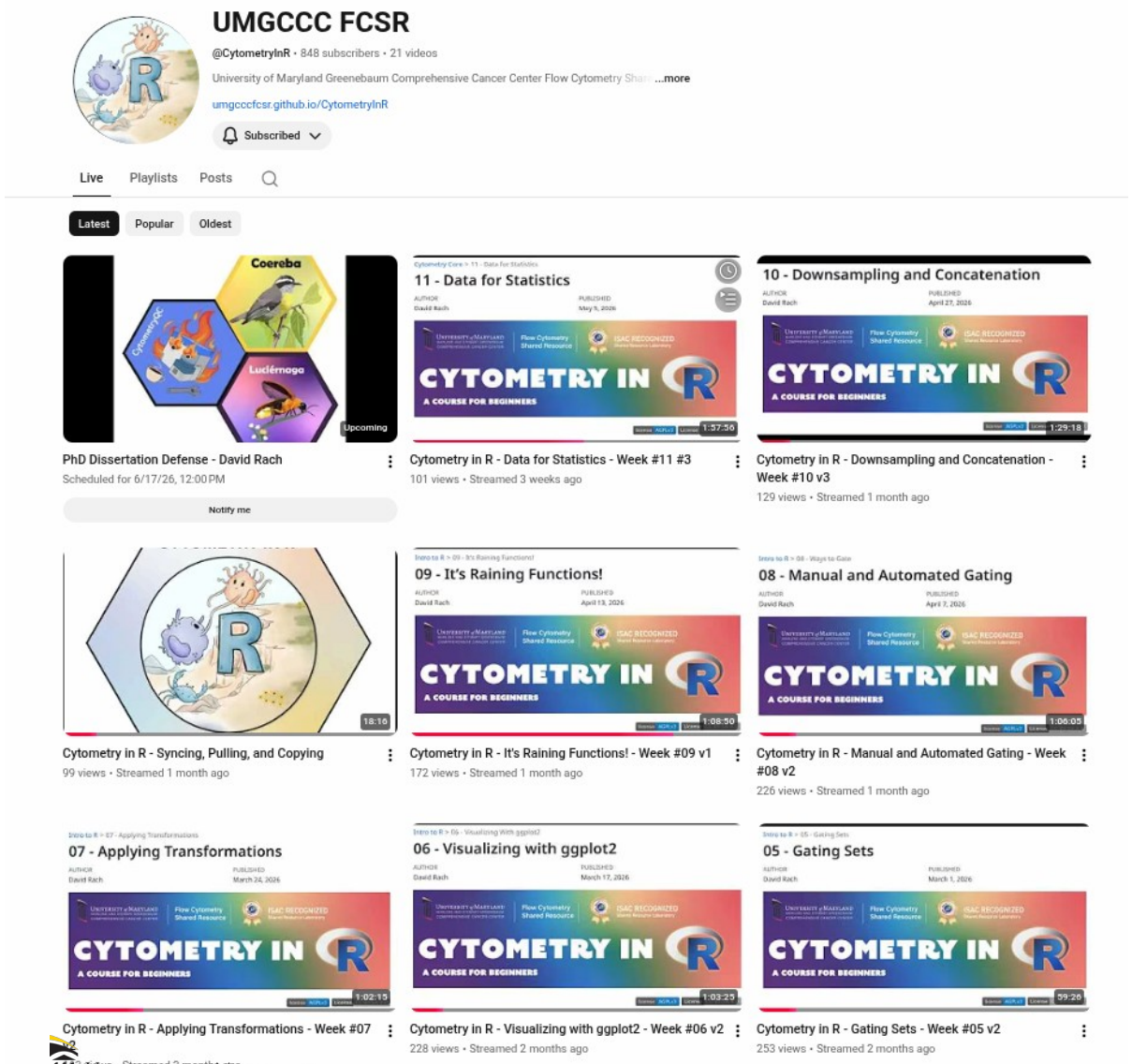
Website



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YouTube

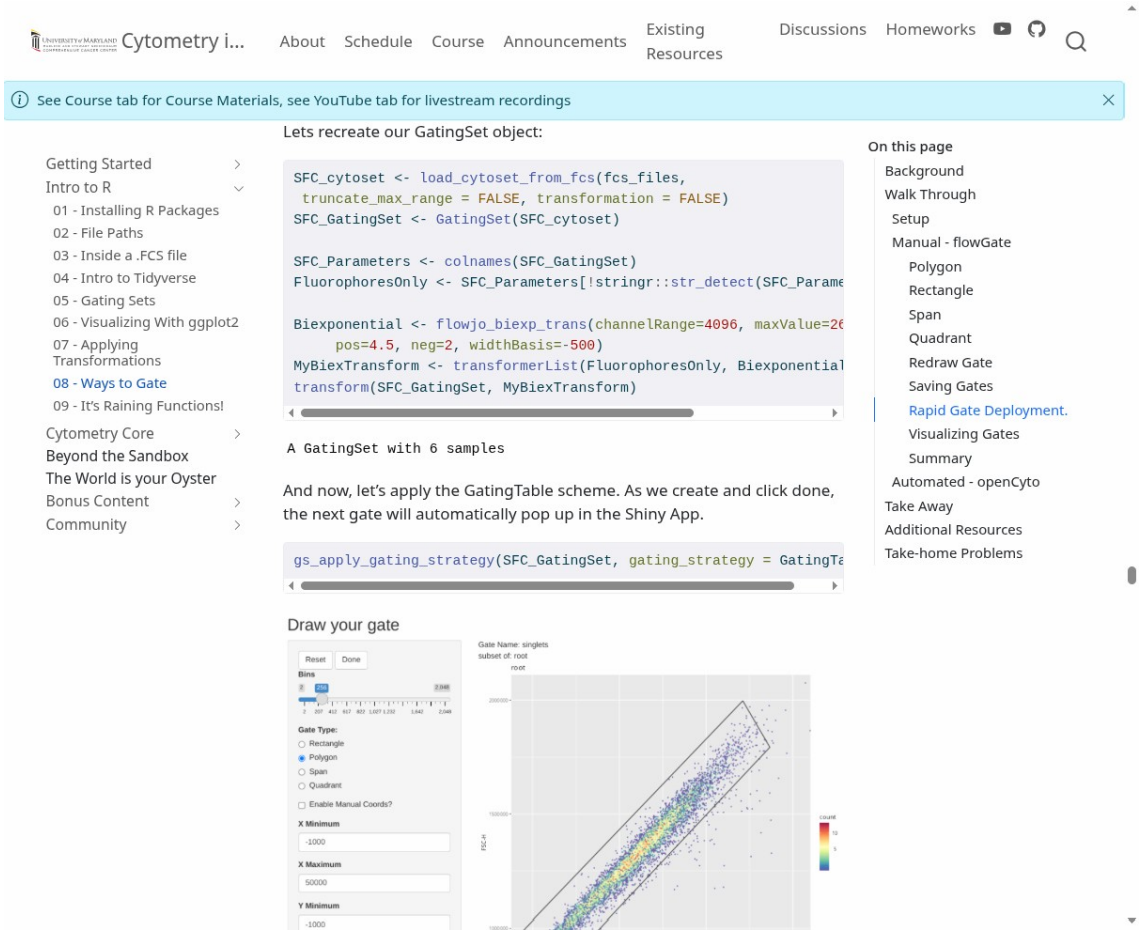


- Three live-streams a week (accommodating different timezones and schedules)
- Recordings available immediately after
- Best of the weekly livestreams added to the playlist, cutting down on editing

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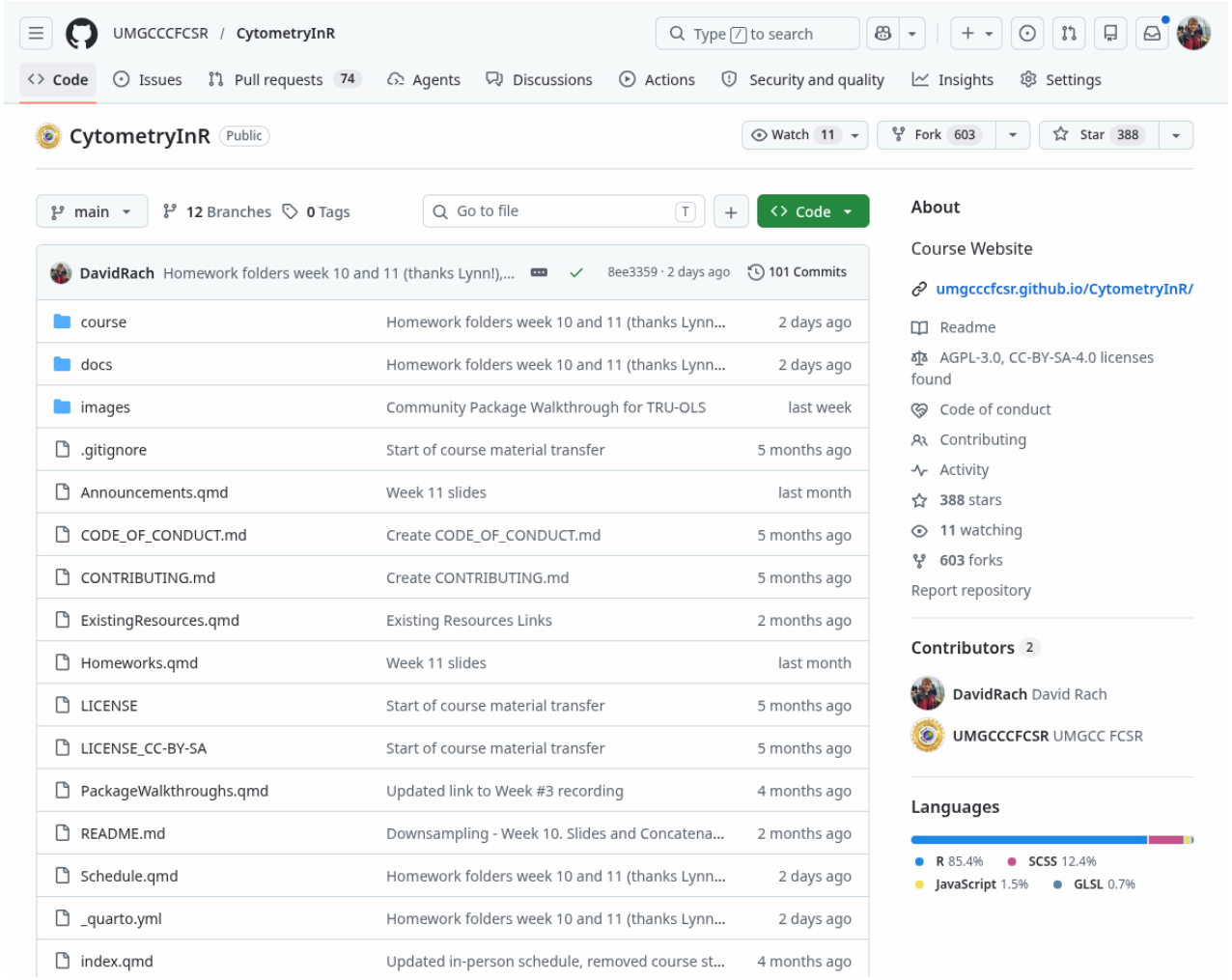
Website

- Created with Quarto, hosted via GitHub pages.
- Code, code outputs, screenshots and explanations
- Searchable
- New weekly material added as a new .qmd file



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GitHub

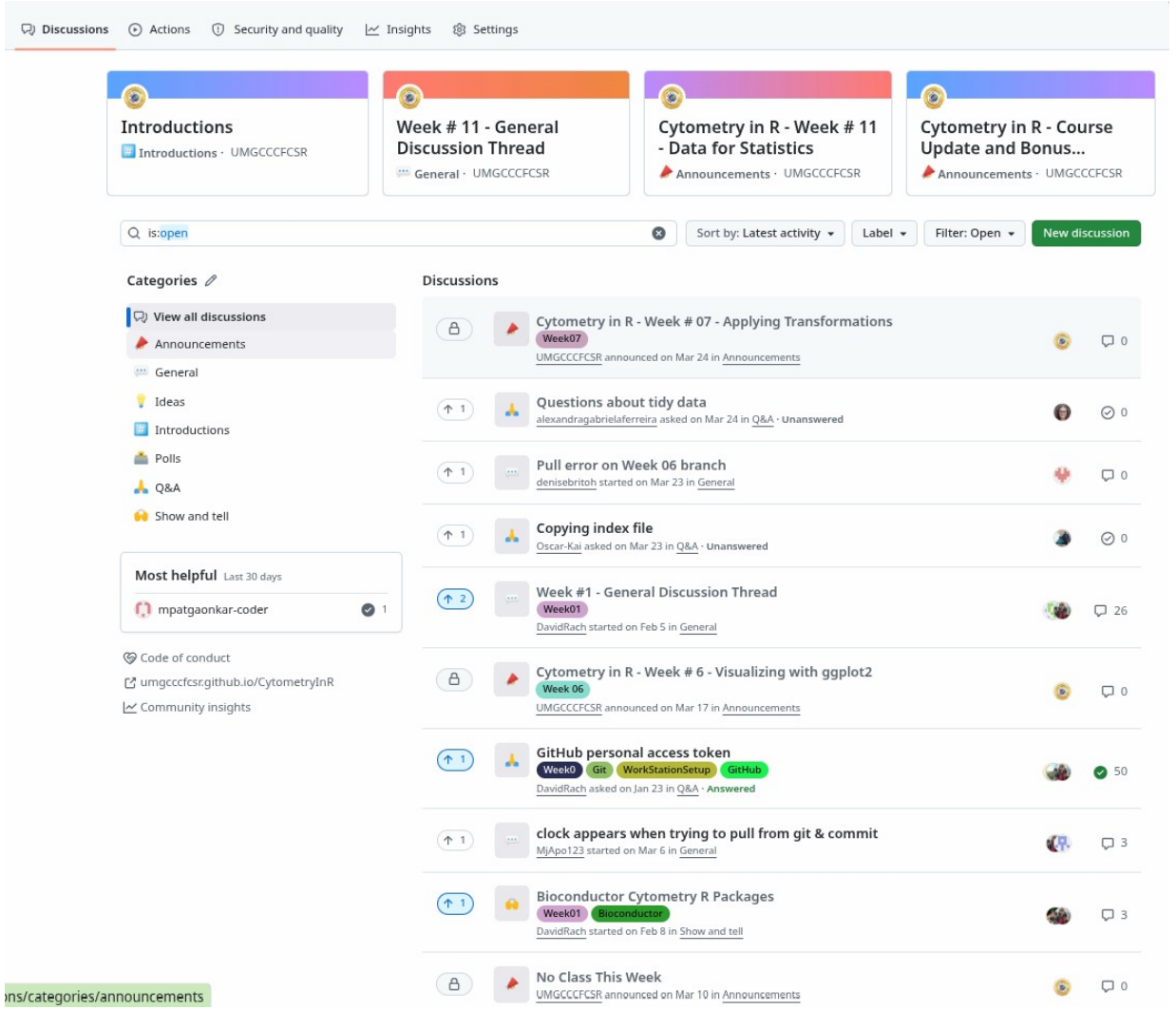


- Code and datasets available via the GitHub repository
- Course participants start course by creating GitHub account and forking the repository
- After Positron is linked to GitHub account, pull locally
- When new material posted each week, sync and pull

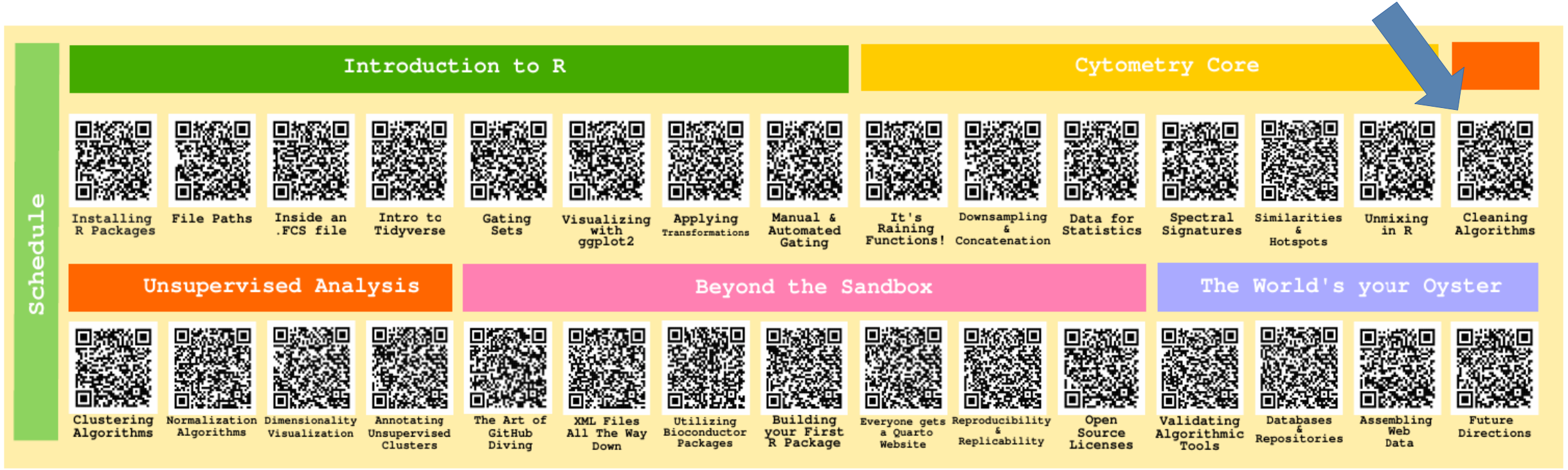
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Discussion Forum

- GitHub Discussions page has been repurposed as a community forum for questions, etc.



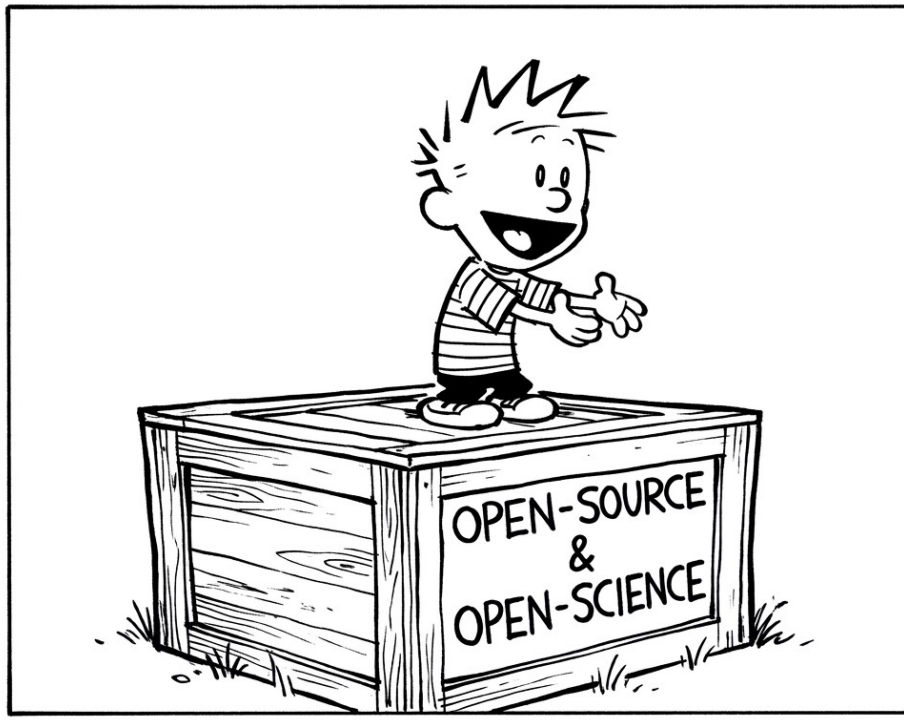
Half-way through this passion project



- Wrapping up Cytometry Core module
- Gathering steam to start on the Unsupervised Analysis section

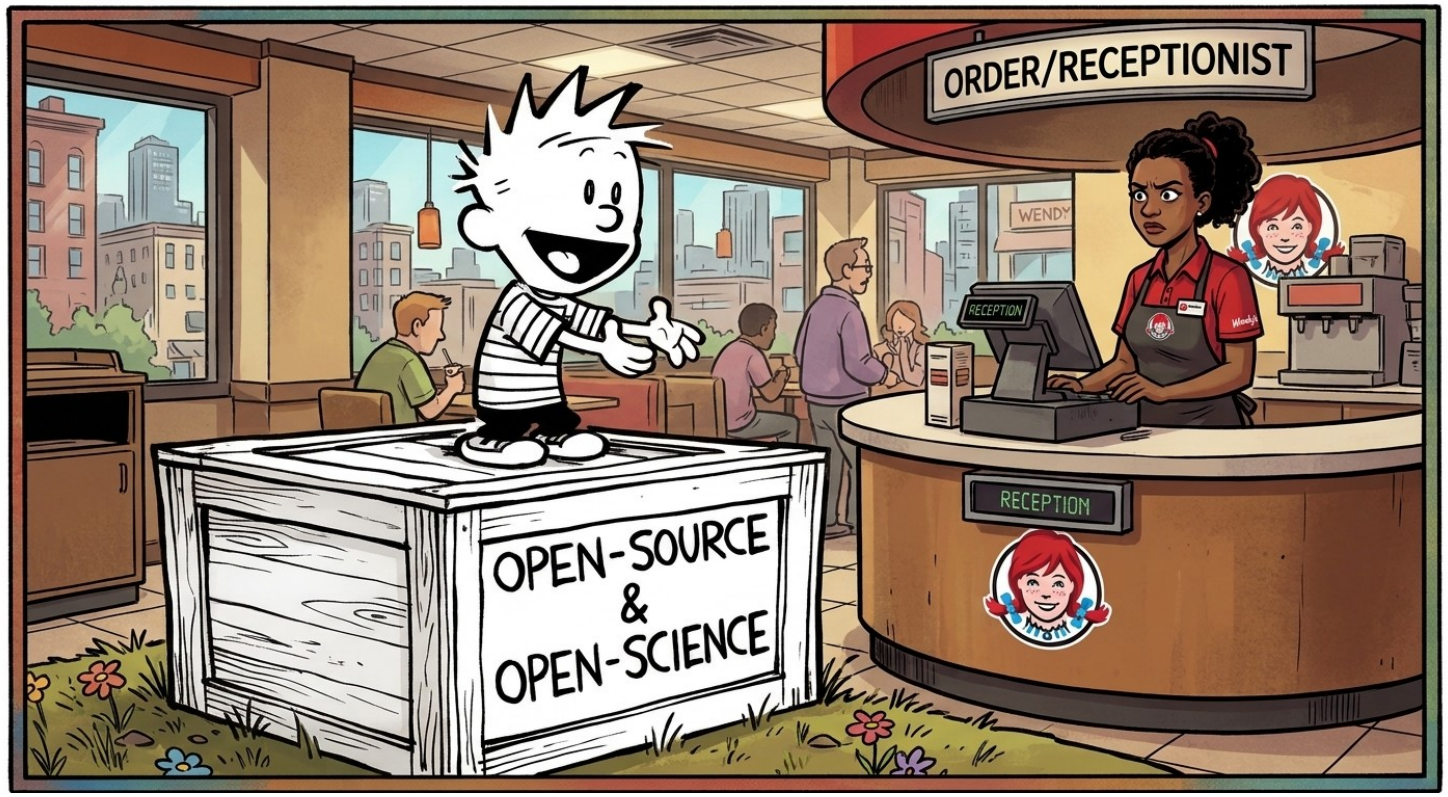
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As scientist, our community is our strength



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Just my two over-caffeinated cents



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Acknowledgments

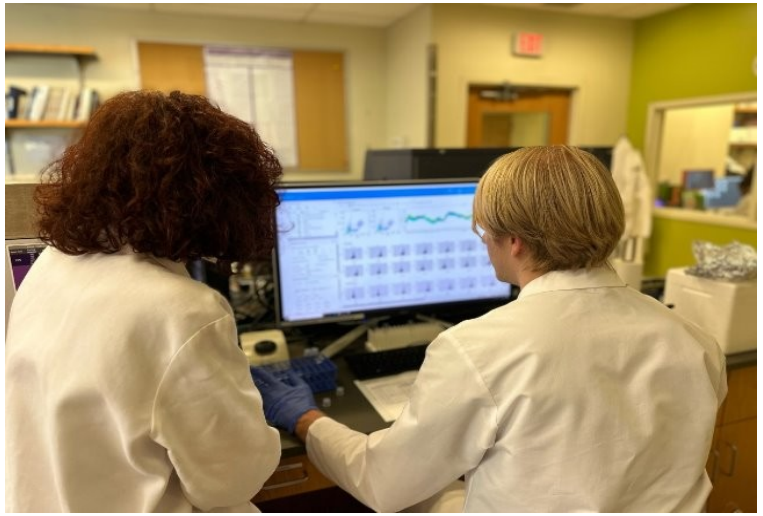
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OF HEALTH SCIENCES

All mothers and their infants, and their communities

All the open-source contributors, developers, and maintainers, that collectively made the tool sets without which all of this would not be feasible

Dr. Cristiana Cairo
Dr. Kirsten E. Lyke
Dr. Xiaoxuan Fan

Dr. Daniela Franco
Dr. Natarajan Ayithan



U01 HD092309

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Cytometry Discord

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Want to collaborate? Reach out!

Hiring? 👀

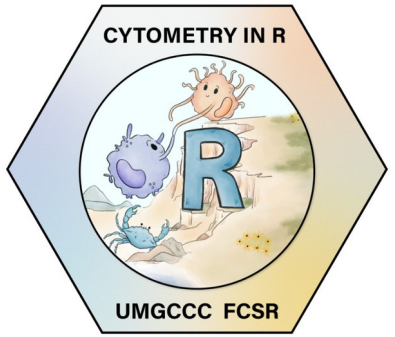
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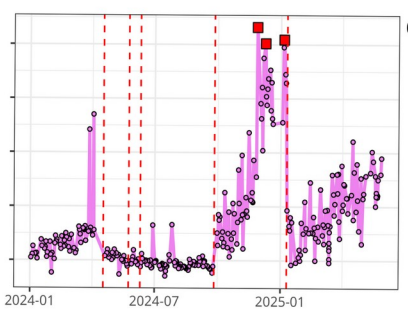
Questions?



Luciernaga



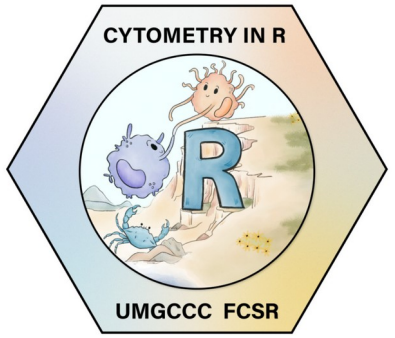
Website - CytometryInR



UMGCCC FCSR
InstrumentQC dashboard



Coereba



YouTube @**CytometryInR**



InstrumentQC How-To

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